Task 10

What is multi-threading programming ??

Multi-threading is a programming technique where a single process runs multiple threads concurrently to perform different tasks simultaneously.

What is thread??

A thread is the smallest unit of execution within a process. Think of it as a separate path of execution that can run code independently.

Multi-threading allows your program to do multiple things at the same time, making it faster and more responsive. Modern C# uses `async/await` and `Task` which are easier than managing threads manually.

Report

1.What is State Management ?

State management is the process of persisting data across multiple requests in a stateless protocol like HTTP.

Type	Description	Example
Client-Side State	Stored on user's browser	Cookies, LocalStorage, SessionStorage
Server-Side State	Stored on server	Session, Cache, Database
Distributed State	Shared across multiple servers	Redis, SQL Server State

2.What is Caching?

Caching is a technique to store frequently accessed data in fast, temporary storage to reduce database load and improve performance.

3.What is Redis?

Redis (Remote Dictionary Server) is an in-memory data structure store used as:
Cache
Database
Message Broker

4.Installing Redis??

1.Local Installation (Windows)

2.Docker (Recommended)

3.Redis Cloud (Free Tier)

5.Adding Redis to ASP.NET Core Project

1.Install NuGet Package
2.Configure in Program.cs
3.Add Connection String

6.Using Redis Cache Code

1.IDistributedCache (Simple)
2.Advanced with Connection Multiplexer

7.Cache Strategies

1.Cache-Aside(Lazy Loading)
2.Write-Through
3.Write-Behind (Write-Back)
4.Cache Eviction Policies
5.Redis Data Structures & Use Cases

Redis is the industry standard for distributed caching. It provides sub-millisecond latency, rich data structures, and built-in expiration. In ASP.NET Core, use `IDistributedCache` for simple scenarios or `StackExchange.Redis` for advanced features.